

3.4. Numerical Examples

In order to compare the three explained discretization methods

- the Galerkin least-squares (GLS) FEM,
- the penalty stabilization (PS) approach and
- the least-squares FEM (LSFEM)

we will now consider numerical examples. Major emphasis is thereby on a possible coupling with a structural field. As explained below, the pressure P takes an important role. The structural forces, which we will compute later on, are based on pressure results.

Altogether we will treat three different test cases in this section. For each test case either results in the literature or an analytic solution exist. The dynamic viscosity is assumed to be one ($\mu = 1$). We will apply second order quadrilaterals with 2×2 integration points as element type. The boundary conditions and all other values are given without a physical unit. Nevertheless, it is straightforward to obtain unit afflicted values out of this unitless values. Thereby, consistency to one unit system is important. E.g., for the SI-system it means, that an assumed length unit in [m] restricts the velocity unit to be [m/s] and the pressure unit to be [N/m²].

We will start with the lid-driven cavity flow. Based on lid-driven cavity, we can formulate an algorithmic estimation for the stabilization parameter τ_m . As mentioned above, we need this parameter for the GLS approach.

Test case two is the channel flow. Hereby, we will consider a flow within a thin and a thick channel. Hidden agenda thereby, is to verify whether the former estimated algorithm for the stabilization parameter (τ_m) works well in a different flow situation. The flow conditions are already close to those of a radial journal bearing, since similar situations arise in radial journal bearings. Next to the smallest gap we have a thin channel and a thick one at the opposite side. For being as close to the radial journal bearing as possible we additionally enforce, that the same volume flow Q is existent in both channels.

The third numerical example has no physical meaning. In this case, we choose a set of equations, which fulfill the Stoke's equations. Based on this test case we will compare the accuracy of GLS, PS and LSFEM.

For each test case we first describe the considered domain and the boundary conditions. Afterwards a selection of simulation results is shown together with an interpretation. At the end of this section we are able to value the different methods. It will be shown, that LSFEM represents the best choice.

3.4.1. Lid-Driven Cavity

The lid-driven cavity represents a widely spread test case for fluid dynamic codes (see e.g. [24] or [51]). Here, we choose an example given in [51]. The domain, as shown in Fig. 3.5, is the unit square. The origin lies in the middle of the domain. The boundary conditions are also shown in Fig. 3.5. The two velocities are known everywhere at the boundary. Additionally, the pressure is known at one point of the boundary.

Results based on GLS, PS and LSFEM will be compared. LSFEM leads to a velocity

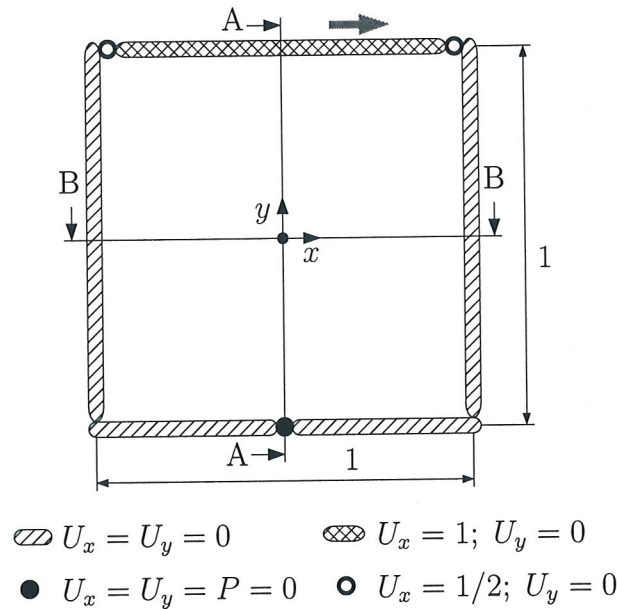


Fig. 3.5. Boundary conditions for the lid-driven cavity

field, as shown in Fig. 3.6. The expected rotating swirl is clearly recognizable.

The pressure field, as given in Fig. 3.7, is also as expected. Results based on the GLS and the PS approach look similar.

A more precise verification is given by the velocity and the pressure distribution along the cross sections A-A and B-B. The positions of these two cross sections are displayed in Fig. 3.5. The results in [51] are presented in the same way. The distributions of the velocity U_x along the cross section A-A are shown in Fig. 3.8. Fig. 3.9 displays the velocity U_y along the horizontal cross section B-B. Finally we can see the pressure distributions along the cross section B-B in Fig. 3.10.

The velocity distributions based on GLS, PS and LSFEM are quite similar. Whereas for the pressure distributions, close to the boundary, this is not the case. As already mentioned, the accuracy of the GLS method depends strongly on a correct chosen stabilization parameter τ_m . For the given discretization, the parameter according to (3.49) would be $\tau_m = 5.2 \cdot 10^{-5}$. Numerical tests showed, however, that $\tau_m = 3.0 \cdot 10^{-5}$ leads to better results. Therefore the parameter determination is modified to

$$\tau_m = \frac{\det(J)}{330} \quad (3.97)$$

This parameter determination proved to be good for different mesh sizes and, therefore, also for different Jacobi determinants. In [51] it is shown, that it is generally possible to gain a better pressure accuracy near the boundary by optimizing the parameter τ_m further. In conclusion we should keep in mind, that the parameter τ_m is proportional to the Jacobi determinant $\det(J)$ in the considered lid-driven cavity.

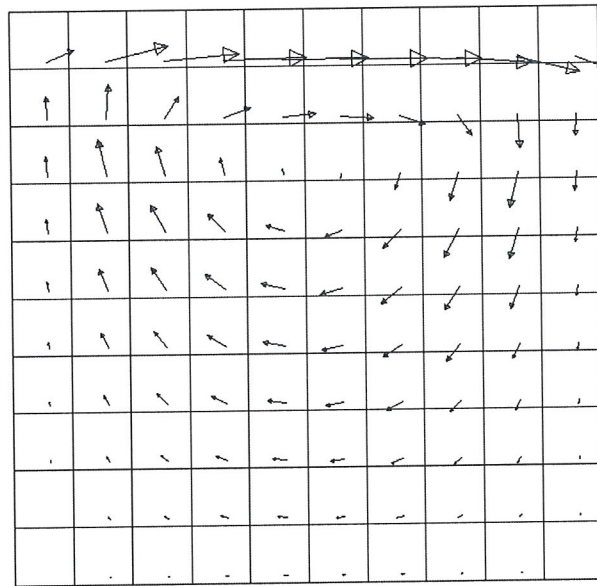


Fig. 3.6. Velocity distribution of the lid-driven cavity based on LSFEM

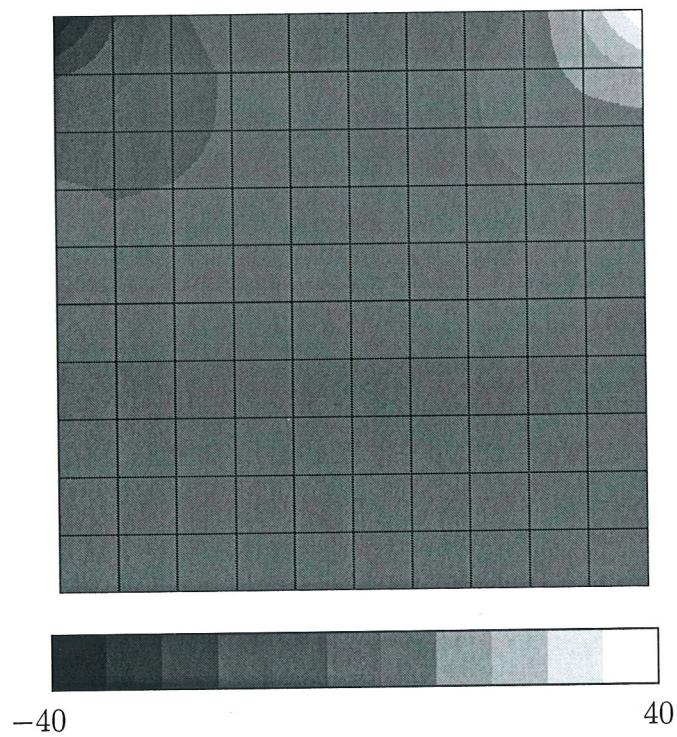
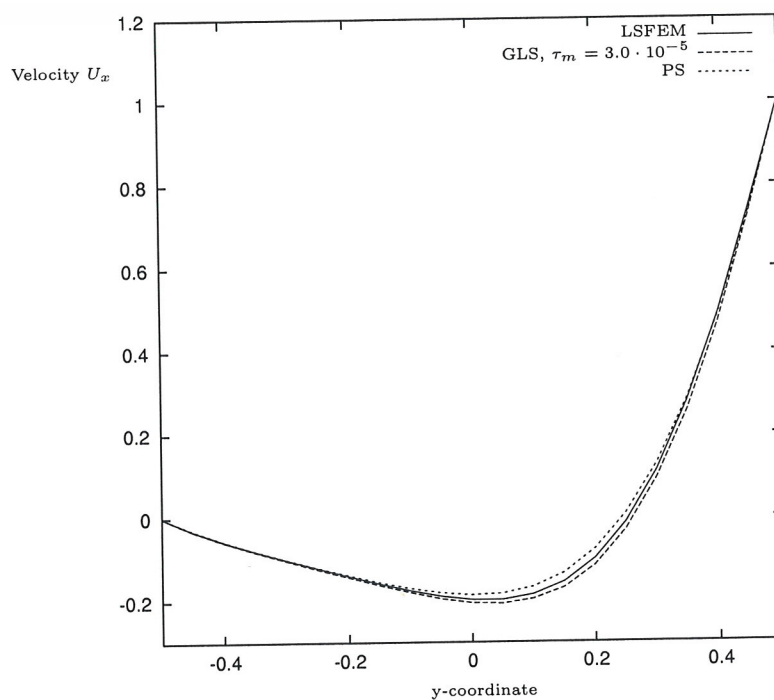
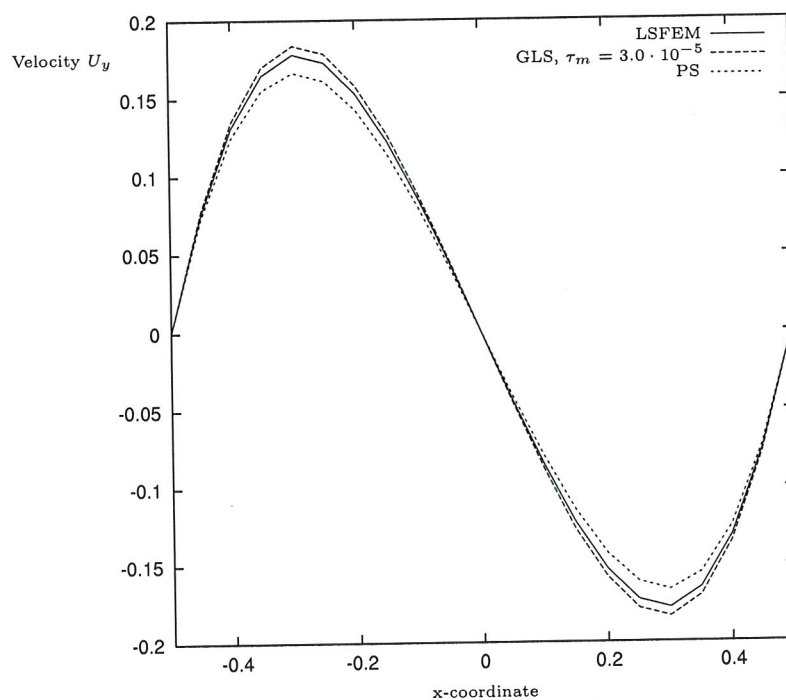


Fig. 3.7. Pressure distribution of the lid-driven cavity based on LSFEM

Fig. 3.8. Velocity U_x along cross section A-AFig. 3.9. Velocity U_y along cross section B-B

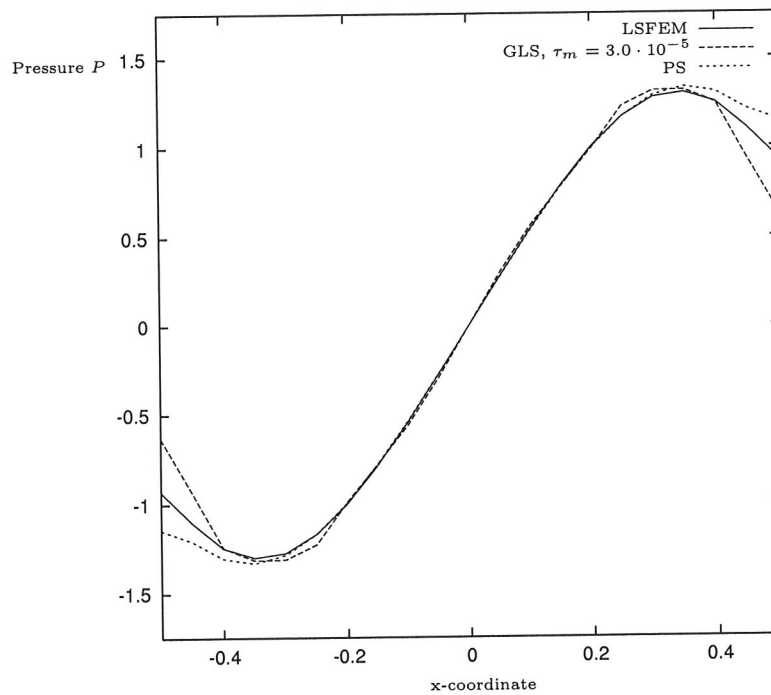


Fig. 3.10. Pressure P along cross section B-B

GLS, PS and LSFEM seem to be suitable to simulate fluid dynamic problems. All methods lead to acceptable results. But, the pressure accuracy near the walls is particularly sensible. The velocity accuracy is thereby not a good measure. However, for a fluid-structure interaction (FSI) we need good pressure results especially close to the boundary.